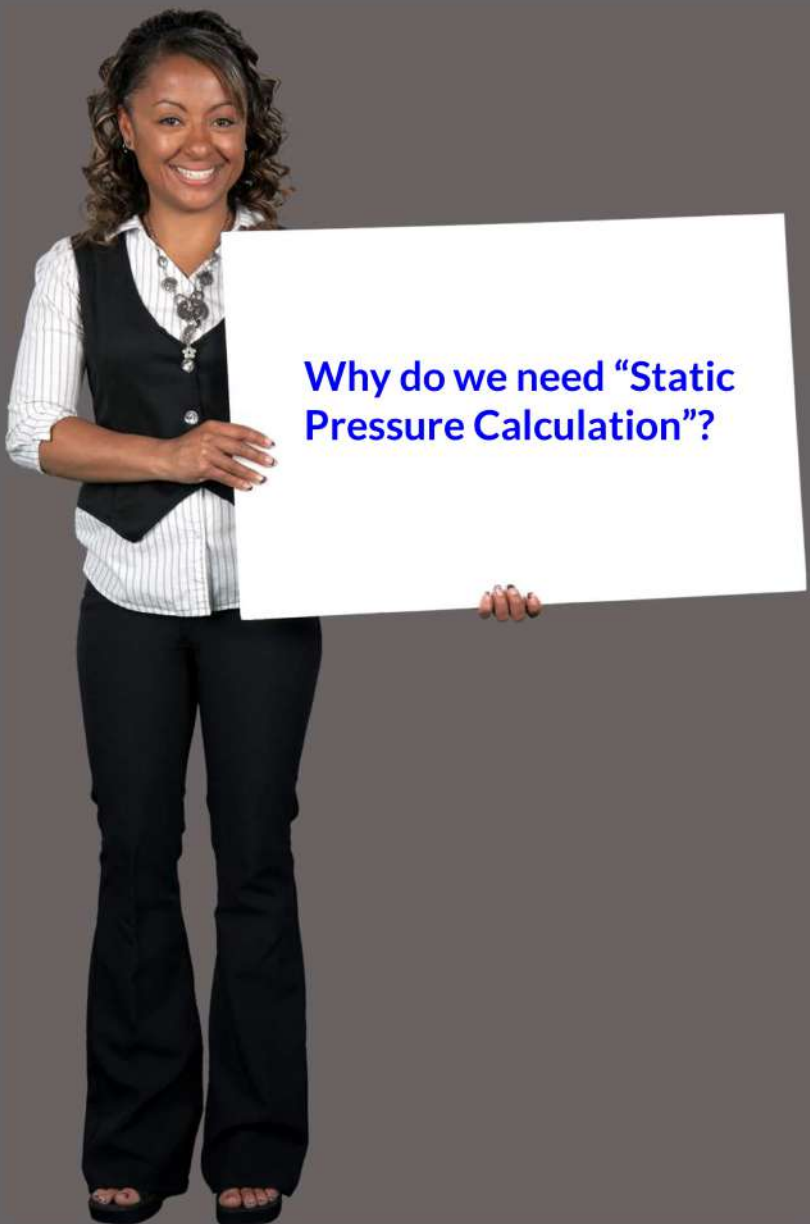


STATIC PRESSURE CALCULATION



A woman with curly hair, wearing a white striped shirt, a dark vest, and dark pants, is smiling and holding a large white sign. The sign has the text "Why do we need 'Static Pressure Calculation'?" written on it in blue.

Why do we need "Static Pressure Calculation"?

- **Static Pressure creates resistance against air movement in the ducts of an ACMV system**
- **Equipment must overcome this pressure to provide heating and cooling**
- **The static pressure and airflow are the two main factors that determine how a fan operates, and also its power consumption**



HERE'S A BREAKDOWN OF KEY CONTEXTS AND CONCEPTS RELATED TO STATIC PRESSURE CALCULATION:

**** Purpose of Static Pressure Calculation:**

- ✓ System Performance
- ✓ Energy Efficiency
- ✓ Comfort

**** Key Concepts:**

- ✓ Static Pressure
- ✓ Dynamic Pressure
- ✓ Total Pressure

**** Components Affecting Static Pressure:**

- ✓ Duct Size and Shape
- ✓ Duct Length
- ✓ Fittings and Accessories
- ✓ Airflow Volume



HERE'S A BREAKDOWN OF KEY CONTEXTS AND CONCEPTS RELATED TO STATIC PRESSURE CALCULATION:

**** Calculation Methods:**

- ✓ Manual Calculation
- ✓ Software Tools

**** Measurement**

- ✓ Static Pressure Measurement

**** Design Considerations:**

- ✓ Duct Design
- ✓ Balancing

**** Standards and Guidelines:**

- ✓ ASHRAE (American Society of Heating, Refrigerating and Air-Conditioning Engineers)
- ✓ SMACNA (Sheet Metal and Air Conditioning Contractors' National Association)

To make ensure that the HVAC system operates efficiently and effectively, by understanding and accurately calculating static pressure, contributing to comfort and energy savings.



OVERVIEW OF GENERAL CALCULATED

Identify Duct System Parameters

- Air Flow Rate
- Duct Dimensions
- Duct Material and Roughness
- Duct Accessories

Pressure Drop for Each Component

- Straight Duct
- Fittings and Accessories

Use Appropriate Formulae

- Duct Area
- Duct Perimeter
- Hydraulic Diameter
- Velocity
- Velocity Pressure
- Reynolds Number
- Friction Factor
- Friction Loss
- Fitting (or) Equipment
- Fitting Number
- Loss Coefficient
- Pressure Loss



Duct Sizing Methods

- ❖ Friction
- ❖ Velocity
- ❖ Equal Friction
- ❖ Static Regain

Duct Size should be limited to a maximum of 4:1. {A duct with an aspect ratio above 4 is much less efficient in use of material and experiences great pressure losses.}



APPROPRIATE FORMULAE

- To Calculate “Duct Area”

Round: $\text{Area} = \pi * (D/12) * L$

Rectangular: $\text{Area} = (W * 2 + H * 2) / 12 * L$

Flat Oval: $\text{Area} = (\pi * H + 2 * W) / 12 * L$

Duct Sizing Methods

- ❖ Friction
- ❖ Velocity
- ❖ Equal Friction
- ❖ Static Regain

Duct Size should be limited to a maximum of 4:1.



APPROPRIATE FORMULAE

- To Calculate “Duct Perimeter”

$$P = (W + H) * 2 , \text{ for rectangular duct}$$

$$P = 2 * \text{Pi} * D/2 , \text{ for round duct}$$

Where:

W = Duct Width, m

H = Duct Height, m

D = Diameter , m



APPROPRIATE FORMULAE

- To Calculate “Hydraulic Diameter”

Darcy-Weisbach equation and Colebrook’s equation

$$D_h = 4A / P$$

Where:

D_h = hydraulic diameter , m

A = duct area , m^2

P = perimeter of cross section, m



APPROPRIATE FORMULAE

- To Calculate “Velocity”

$$V = Q / A$$

Where:

V = velocity, m/s

Q = Air Flow, CMS

A = Area, m²



APPROPRIATE FORMULAE

- To Calculate “Velocity Pressure”

$$P_v = \rho * v^2 / 2$$

Where:

P_v = Velocity Pressure, Pa

ρ = density, kg/ m³

V = Velocity, m/s



APPROPRIATE FORMULAE

- To Calculate “Reynolds Number”

$$Re = 66.4 * Dh * V$$

Where:

Re = Reynolds Number

Dh = Hydraulic Diameter, mm

V = Velocity, m/s



APPROPRIATE FORMULAE

- To Calculate “Friction Factor”

$$f' = 0.11[(\varepsilon/D_h) + (68/Re)]^{0.25}$$

Where:

f' = Friction Factor

ε = Duct roughness factor, mm

D_h = Hydraulic diameter, mm

Re = Reynolds Number

If :

$f' \geq 0.018$, $f = f'$

$f' < 0.018$, $f = 0.85f' + 0.0028$

*Duct roughness factor for GI duct = 0.15

*Duct roughness factor for Concrete duct = 3



APPROPRIATE FORMULAE

- To Calculate “Friction Loss”

$$\Delta P_f = 1000 * f * P_v / D_h$$

Where:

ΔP_f = Friction Loss, Pa/m

f = Friction factor

P_v = Velocity pressure, Pa

D_h = Hydraulic diameter, mm



- **Fitting (or) Equipment** : get from ducting design layout
- **Fitting Number** : get from ASHRAE Table
- **Loss Coefficient “Co”** : get from ASHRAE Table
- **To Calculate “Pressure Loss”**

Pressure Loss (Pa) = $P_v * Co$, for fitting

Pressure Loss (Pa) = Duct Length * ΔP_f , for ducting

Where:

ΔP_f = Friction Loss, Pa/m

Co = Loss Coefficient

P_v = Velocity Pressure, Pa



ASHRAE Duct Fitting Database

File Utilities Project Administrator Help

CR3-1

Supply

- Round
 - Plenums
 - Exits
 - Transitions
 - Junctions, Diverging
 - Fan & System Interactions
- Rectangular
- Flat Oval

Common

- Round
 - Straight Duct
 - Straight Duct, Flexible
 - Elbows
 - Obstructions
 - Duct-Mounted Equipment
 - Dampers
- Rectangular
 - Straight Duct
 - Elbows
 - Mitered
 - Without Vanes
 - Double-Thickness Vanes
 - 1 1/2-in. Vane Spacing
 - 2 1/8-in. Vane Spacing
 - 3 1/4-in. Vane Spacing
 - Single-Thickness Vanes
 - 1 1/2-in. Vane Spacing
 - 3 1/4-in. Vane Spacing
 - Other Elbows
 - Smooth Radius
 - Without Vanes
 - One Splitter Vane
 - Two Splitter Vanes
 - Three Splitter Vanes
 - Obstructions
 - Duct-Mounted Equipment
 - Dampers
 - Flat Oval

CR3-1 Elbow, Smooth Radius without Vanes (Idelchik 1986, Diagram 6-1)

Input		Output	
Width (W, mm)	300	Velocity (V _o , m/s)	7.8
Height (H, mm)	300	Vel Pres at V _o (P _v , Pa)	36
Radius Ratio (r/W)	1.5	Loss Coefficient (C _o)	0.17
Angle (Theta, deg.)	90	Pressure Loss (Pa)	6
Flow Rate (Q, L/s)	700		

CR3-1 Elbow, Smooth Radius without Vanes (Idelchik 1986, Diagram 6-1)

$C_o = K C_p$

where
K = angle factor

W x H

Common\Rectangular\Elbows\Smooth Radius\Without Vanes

ASHRAE duct fitting database software

ASHRAE duct fitting database includes the loss coefficients for all 235 duct fittings.

